Injectable anabolic-androgenic steroids (AAS): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-09.

Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for Injectable anabolic-androgenic steroids (AAS) (Peptides, SARMs, and Research Chemicals), evaluated on the finished product as sold. Every analyte is held to the ICH Q3D(R2) elemental-impurity limit for the parenteral route (the harmonized pharmaceutical standard, enforceable via USP <232>), measured on every lot by ICP-MS. Certification is heavy-metal safety only and does not endorse the efficacy or legality of these products.

1 | How every number on this standard is set

HMTc Method v2.1. Lead, cadmium, total arsenic, total mercury, nickel, tin, and total chromium are read across from the ICH Q3D(R2) Table A.2.2 Option 1 parenteral concentration limits. Aluminium is read across from the FDA large-volume parenteral limit, 21 CFR 201.323, at 25 ug/kg. Inorganic arsenic, methylmercury, and hexavalent chromium do not carry a separate numeric ceiling. Each is capped at its parent total and tested on every lot.

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	500	read-across (ICH)

Cadmium (Cd)	200	read-across (ICH)
Total arsenic (tAs)	1,500	read-across (ICH)
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	300	read-across (ICH)
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	2,000	read-across (ICH)
Tin (Sn)	60,000	read-across (ICH)
Aluminium (Al)	25	read-across (US FDA)
Total chromium (Cr)	110,000	read-across (ICH)
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	500	read-across (ICH); parenteral (injectable), nasal, and sublingual drug products
Cadmium (Cd)	200	read-across (ICH); parenteral (injectable), nasal, and sublingual drug products
Total arsenic (tAs)	1,500	read-across (ICH); parenteral (injectable), nasal, and sublingual drug products
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	300	read-across (ICH); parenteral (injectable), nasal, and sublingual drug products
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	2,000	read-across (ICH); parenteral (injectable), nasal, and sublingual drug products
Tin (Sn)	60,000	read-across (ICH); parenteral (injectable), nasal, and sublingual drug products

Aluminium (Al)	25	read-across (US FDA); parenteral (injectable) drug products; FDA 21 CFR 201.323 (25 ug/L approx 25 ug/kg)
Total chromium (Cr)	110,000	read-across (ICH); parenteral (injectable), nasal, and sublingual drug products
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

4 | Cite this standard

Pendergrass, K. (2026). *Injectable anabolic-androgenic steroids (AAS): HMTc standard* (Standard No. 10.83138/hmtc.std.26.aas-injectable). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.26.aas-injectable

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References

[1] ICH
Binds Pb, Cd, tAs, tHg, Ni, Cr, Sn. parenteral (injectable), nasal, and sublingual drug products.
READ-ACROSS

[2] US FDA
Binds Al. parenteral (injectable) drug products; FDA 21 CFR 201.323 (25 ug/L approx 25 ug/kg).
READ-ACROSS

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

This standard is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this standard draws on are listed in full in the references above and on the standard page at https://heavymetalcertified.com/standards/aas-injectable. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-09. The full dated record is maintained at https://heavymetalcertified.com/standards/aas-injectable.